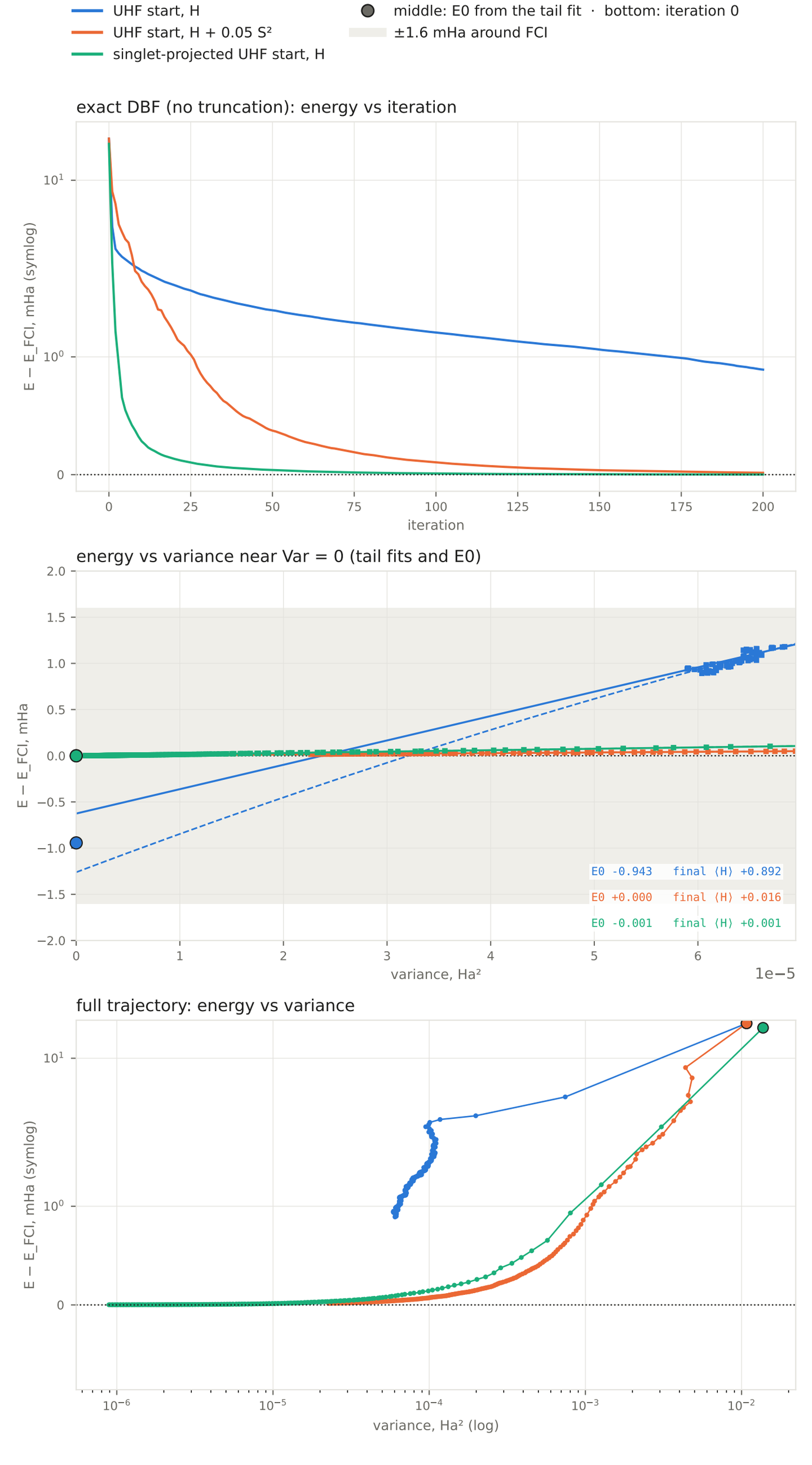LNO, R = 2.3 Å, frozen core (16 qubits)